# Essays

|!|BCEEC0050000001|!|

Account for the factors that affect the rate of enzymatic reactions. (12 marks)

##

| Content | Description | Mark |
|---|---|---|
| Temperature | - At low temperatures, most enzymes are inactive.<br>- As the temperature rises, the substrate and enzyme molecules have more kinetic energy and move faster. The substrate and enzyme molecules collide with one another and enzyme-substrate complexes are more likely to form.<br>- The rate of an enzymatic reaction reaches its maximum at the optimum temperature.<br>- When the temperature increases beyond the optimum temperature, the shapes of the active sites of the enzymes start to change.<br>- Enzymes are denatured at very high temperatures.<br>- The loss of catalytic function of enzymes at high temperatures is irreversible. | (Max. 4) |
| pH | - Enzymes work best at their optimum pH.<br>- A change in pH alters the shape of the enzyme molecules and their active sites.<br>- Most enzymes are denatured at extreme pH values. | |
| Inhibitors | - Inhibitors (e.g. cyanide, heavy metal ions) bind to enzymes and decrease the rate of enzymatic reactions. | |
| Enzyme and substrate concentration | - The higher the substrate concentration, the higher is the reaction rate until the rate of reaction is limited by the enzyme concentration.<br>- When the rate of reaction is limited by the enzyme concentration, the rate of reaction can be increased if more enzyme molecules are added. | (Max. 5) |
| **Communication** | | (Max. 3) |

(12 marks)

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| Skills assessed | Marks |
|---|---|
| Understanding basic concepts | 9 |
| Communication | 3 |

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|!|BCEEB0050000002|!| ★ **Cross topic**

Discuss the nature of enzymes and the importance of enzymes to humans. (11 marks)

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| Content | Description | Mark |
|---|---|---|
| Nature of enzymes | - protein structure<br>- act as biological catalysts / speed up chemical reactions<br>- reusable<br>- specific in action<br>- form enzyme-substrate complex with substrate<br>- activity affected by temperature, pH and inhibitors | (Max. 3) |
| Importance to humans (physiological) | - catalyse metabolic reactions (any two of the following examples + explanation)<br>• digestion (breakdown of large organic food substances)<br>• respiration (breakdown of glucose)<br>• assimilation (synthesis of organic food substances)<br>• detoxification (get rid of hydrogen peroxide by catalase)<br>(Or other relevant examples) | (Max. 3) |
| Importance to humans (everyday life) | - Meat tenderizers<br>- Clarifying fruit juice<br>- Removing lactose from milk<br>- Making baby food<br>- Making contact lens cleaner<br>- Making biological washing powders<br>- Making stonewashed jeans<br>- Making glucose test strips<br>- Production of biofuels<br>(Or other relevant examples) | (Max. 2) |
| **Communication** | | (Max. 3) |

(11 marks)

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| Skills assessed | Marks |
|---|---|
| Understanding basic concepts | 8 |
| Communication | 3 |

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